

FROZEN-MSE DATA DIAGNOSTIC

We evaluate existing, late-trained MSE policies on demonstration observations. These are training-data fit diagnostics: the episodes were used to train the policies. Separate episode groups are used only to validate scale estimates.

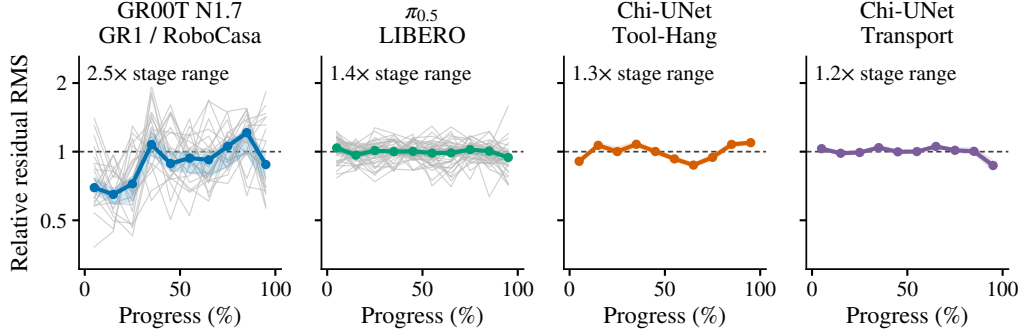


Figure 1: **Residual scale across demonstration progress.** Frozen MSE policies are evaluated on demonstration observations. Each task’s continuous-action residual RMS is normalized by its pooled RMS with equal weighting of available progress bins. Gray lines show individual tasks; colored lines show their median, with episode-bootstrap 95% confidence bands. Annotations give the median within-task maximum-to-minimum stage ratio. Binary grippers are excluded; GR1 hand joints are retained. Progress bins are a stage proxy, not semantic phase labels.

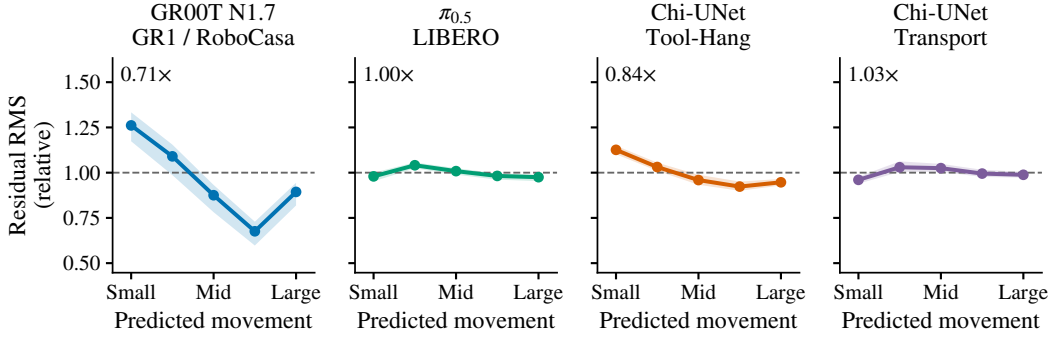


Figure 2: **Movement size and MSE fitting residuals.** Observations are grouped into within-task quintiles of predicted movement magnitude after undoing action normalization. The vertical axis shows continuous-action residual RMS relative to each task’s pooled RMS; tasks are equally weighted. Bands are episode-bootstrap 95% confidence intervals. Annotations compare the largest- and smallest-movement groups. Target actions are not used to define the groups.

INDEPENDENT-EPISODE CHECK

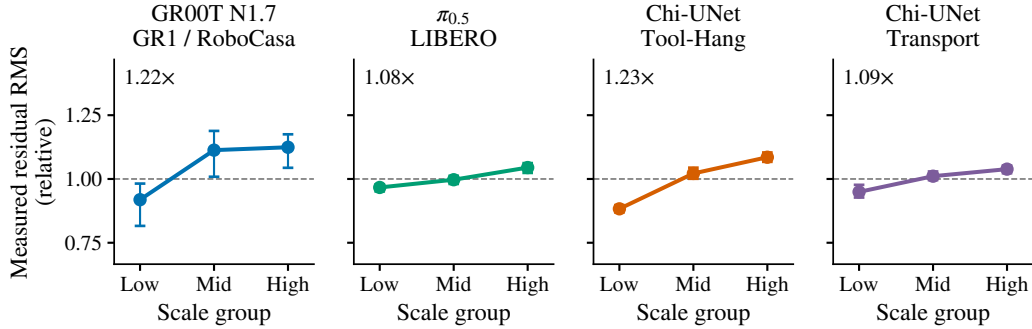


Figure 3: An exploratory observation-dependent scale check. Within each task, nearest neighbors from a separate episode group estimate residual energy using proprioception. Query observations are grouped by that predicted scale, without using their target actions or residuals. The plotted query residual RMS is normalized within task; tasks are equally weighted. Error bars are episode-bootstrap 95% intervals conditional on the fitted scale predictor.

Interpretation. Task-dependent precision is a hypothesis about the demonstration process, not a consequence of unequal action magnitudes. Signal-dependent motor noise and task-constrained variability provide relevant motivation (Harris & Wolpert, 1998; Todorov & Jordan, 2002). Fitting residuals also contain model error and need not equal the underlying demonstration noise. No Gaussian-mixture or discrete-scale claim follows from these plots.

REFERENCES

- Christopher M. Harris and Daniel M. Wolpert. Signal-dependent noise determines motor planning. *Nature*, 394(6695):780–784, 1998. doi: 10.1038/29528.
- Emanuel Todorov and Michael I. Jordan. Optimal feedback control as a theory of motor coordination. *Nature Neuroscience*, 5(11):1226–1235, 2002. doi: 10.1038/nn963.